

Model Catalog

This file lists every model family and saved bundle used in the project, along with its purpose in the screening workflow.

Text

- text_transformer: primary transformer-based text scorer for narrative screening; used when the local transformer bundle is available.
- text: classical text scorer and fallback bundle; used when the transformer bundle is unavailable or as a stability backup.

Audio

- audio: primary acoustic screening bundle; the main audio scorer for mood-heavy and speech-derived signals.
- audio_spectrogram: secondary audio branch trained on spectrograms; used as a stabilizer when available.
- audio_sequence: secondary audio branch trained on chunk sequences; used as a fallback stabilizer until it improves further.

Image

- image_dl: primary image scorer; the current CNN-based image bundle with stronger augmentation and better diversity.
- image: classical image bundle; used as a fallback and stabilizer for face-derived features.

Comorbidity

- comorbidity: classifier-chain ensemble for joint label prediction; used to estimate co-occurring domain patterns and now blend with other modalities.

Supporting Runtime Components

These are not final decision models, but they support the screening stack:

- questionnaire scoring: structured symptom scoring from the assessment workspace.
- passive biomarkers: typing and phone-camera-derived adjunct signals for low-hardware screening support.
- heuristic NLP features: sentiment, emotion, distress phrases, and safety-language cues that keep the app usable when transcription fails.

Practical Priority Order

The runtime currently prefers:

1. text_transformer over text
2. audio over audio_spectrogram and audio_sequence
3. image_dl over image
4. comorbidity after upstream modality consensus is available

Where These Models Are Used

- Preview scoring in the dashboard

- Saved assessment scoring
- Analytics Hub model statistics
- PDF report summaries
- Quality-check and trajectory views
- Training and retraining workflows

Related Files

- docs/MODEL_COMPARISON_MATRIX.md (MODEL_COMPARISON_MATRIX.md)
- docs/MULTIMODAL_FUSION.md (MULTIMODAL_FUSION.md)
- docs/DEEP_LEARNING_MODEL_GUIDE.md (DEEP_LEARNING_MODEL_GUIDE.md)
- README.md (./README.md)